Attributed Scattering Center Guided Network Based on Omnidirectional Subaperture Division for SAR Target Detection

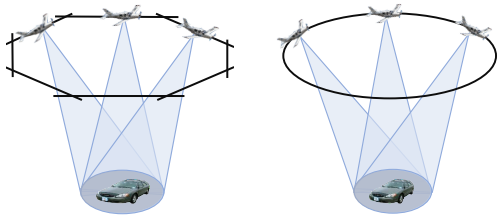
Di Wang, Yongping Song, Leping Chen, and Daoxiang An, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) multiview observations can acquire omnidirectional scattering characteristics of targets, providing richer information for target detection and recognition. Deep learning has been widely applied to SAR image interpretation in recent years. However, the sensitivity of SAR images to imaging parameters and the limited training samples pose challenges in applying deep learning to SAR target detection. Therefore, this article first designs an SAR omnidirectional subaperture division data enhancement method with different aspect accumulation angles, multilook numbers, and subaperture overlap angles based on SAR imaging parameters and the original echo. Meanwhile, an omnidirectional SAR target detection dataset called SAR-Vehicle-Det is constructed using this subaperture division method to evaluate the model performance. Then, this article proposes a novel network attributed scattering center (ASC)-U^{2}Det based on ASC guidance for SAR target detection. The prediction is divided into the ASC reconstruction image branch and the target detection branch. The reconstruction branch uses ASC reconstruction mask maps to guide the network to extract the SAR target scattering features at different aspect angles, suppress the background clutter interference, and improve the detection accuracy of the target detection branch. Finally, this article also evaluates the impact of imaging parameters with different subaperture accumulation angles and multilook numbers on the network detection performance. Experiments on the SAR-Vehicle-Det and miniSAR datasets show that the proposed ASC-U^{2}Det outperforms many existing target detection algorithms.**

***Index Terms*— Attributed scattering center (ASC), subaperture, synthetic aperture radar (SAR), vehicle target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active sensor operating in the microwave band. Due to the ability to work in all-weather and all-day, SAR has been widely used in military reconnaissance, geological exploration, and image reconstruction [1], [2], [3]. With the improvement of radar resolution, the ability of models to describe the scattering characteristics of SAR targets has been enhanced. Targets with anisotropic geometries have different scattering characteristics at different aspect angles, providing richer information for target detection and recognition, and existing works have demonstrated that using SAR images from different aspect angles can improve target recognition accuracy [4], [5].



(a) (b)

Fig. 1. Omnidirectional imaging schematic. (a) Schematic of the LSAR flight path. (b) Schematic of the CSAR flight path.

To obtain SAR images of targets with different aspect angles, SAR systems are required to observe the target area for a long time with multiple angles. Most SAR systems are divided into linear SAR (LSAR) and circular SAR (CSAR) [6]. LSAR platforms always fly in a straight trajectory and can only capture the backscattering characteristics of the target within a limited angle [7]. Therefore, to obtain the omnidirectional scattering characteristics of targets, multiple flight paths need to be designed, as shown in Fig. 1(a). In contrast, CSAR platforms fly in a circle around the observation scene, as shown in Fig. 1(b). The radar platforms perform 360° observation of the target area, allowing CSAR to capture the omnidirectional scattering characteristics of the target in a single flight path [8]. The sidelobes of strong scattering points in different aspect angles can affect target detection and recognition [9]. Teng et al. [10] analyzed the anisotropic scattering of CSAR targets and proposed aspect entropy (AE) for the classification of typical targets, such as dihedral and trihedral. Xue et al. [11] analyzed the variation of CSAR target features on subaperture images using multiaperture polarimetric entropy and were able to classify isotropic and anisotropic targets. Pei et al. [4] have demonstrated that the multiangle features of targets can enhance target recognition performance. From Fig. 2, it is evident that targets exhibit typical anisotropic scattering characteristics at different aspect angles, indicating that the corresponding subaperture images are also different. Whether using LSAR or CSAR, the subaperture division is related to the selection of SAR imaging aspect accumulation angle and

Received 10 October 2024; revised 29 December 2024 and 5 February 2025; accepted 8 March 2025. Date of publication 12 March 2025; date of current version 28 March 2025. This work was supported by the National Natural Science Foundation of China under Grant 62271492. *(Corresponding author: Daoxiang An.)*

The authors are with the College of Electronic Science Technology, National University of Defense Technology, Changsha 410073, China (e-mail: wdnudt@163.com; sypopqjkl@163.com; gfkdclp@126.com; daoxiangan@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3550867

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

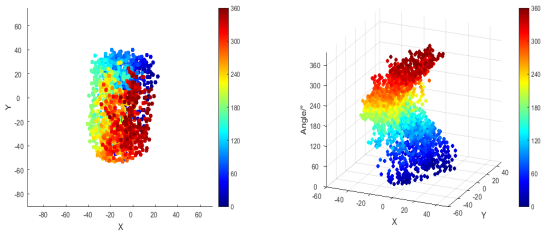


Fig. 2. (a) 2-D map of the scattering feature locations of the vehicle target at different aspect angles. (b) 3-D map of vehicle target scattering feature locations at different aspect angles.

multilook numbers [12], so different imaging parameters bring certain challenges to the SAR image interpretation and dataset construction. Therefore, this article intends to obtain the omnidirectional scattering characteristics of SAR targets through subaperture division right from the dataset construction stage.

In recent years, researchers have created several SAR datasets [13], [14], [15], [16] to promote the development of SAR target detection. Most of these datasets are sea backgrounds with ship targets. In contrast, there are few publicly available ground vehicle target detection datasets, such as miniSAR [17], which also face the problem of limited data. To expand SAR vehicle datasets, methods such as generative adversarial networks (GANs) [18] and computer-aided design (CAD) [19] for generating simulated data have been proposed. Sun et al. [20] utilized the geometric imaging information of vehicle targets and combined it with GAN to generate SAR images with different aspect angles. Liu et al. [21] have generated a synthetic SAR vehicle detection dataset Mix-MSTAR by linearly combining target masks and clutter and improving background migration. However, due to the complex imaging mechanism of SAR images, there are still differences between these synthetic datasets and real scenes. Several studies have employed subaperture decomposition techniques for data augmentation [22], [23], [24], focusing on converting single-channel SAR images into multichannels. In practice, the fast Fourier transform (FFT) is performed on the SAR image along the aspect dimension to obtain the Doppler spectrum, and the spectrum is divided into multiple subspectrums. Subsequently, the inverse FFT (IFFT) is conducted to revert to the image domain. However, this method is based on the image domain, where the aspect angle is fixed, resulting in limited image information obtained from the decomposition. The division of subapertures in the echo domain is commonly used for real-time SAR imaging [25] and 3-D reconstruction [26], and the diversity of subaperture divisions directly affects the target scattering characteristics. Currently, few studies consider the construction of SAR datasets from the perspective of imaging parameters. Therefore, this article combines the advantage of SAR continuous multiview observations, directly dividing the subapertures with different aspect accumulation angles, multilook numbers, and overlap angles based on the original echo domain, and reimaging all the divided pulse echoes to obtain subaperture images. This approach aims to generate a large number of diverse samples within a limited number of observations and then construct the omnidirectional SAR vehicle target detection dataset of the real scene to validate the model performance.

For the usage of SAR target detection and recognition datasets, most researchers currently adopt data-driven deep learning methods and have achieved good performance in terms of speed and accuracy [27], [28], [29]. Considering the practical application of SAR, the interpretability of deep networks has also received extensive attention from researchers [30]. Most scholars fuse manually extracted SAR physical scattering features such as point scattering features and attributed scattering center (ASC) features with deep semantic features extracted by deep networks to improve the interpretability of SAR classification [31], [32], [33]. However, most of these networks divide the scattering feature maps and raw SAR images into two branches at the input side to extract features separately and only perform feature-level fusion before the network output. In this article, we consider using physics knowledge to guide the networks to extract features. Therefore, inspired by semantic segmentation [34] networks, semantic segmentation masks assign a semantic label to each pixel, which can help the network achieve a more detailed and accurate understanding of the target. Zhao et al. [35] used Gaussian mask maps to assist in SAR ship target localization. Wang et al. [36] similarly used Gaussian heat maps to guide attention networks and improve target detection accuracy. However, the mask maps that they used are all Gaussian distributions, and for SAR images, the actual target features are not simple Gaussian distributions. Thus, in this article, the ASC mask maps are constructed and used as supervisory information at the output end to guide the network to extract scattering features, thereby improving the network's interpretability.

In summary, the application of deep learning to SAR vehicle target detection encounters several challenges: the impact of different SAR aspect accumulation angles and multilook imaging parameters on subaperture imaging results, the few publicly available SAR vehicle target detection datasets, and the lack of interpretability in deep networks. To address these issues, this article first designs the SAR subaperture division data enhancement method with different aspect accumulation angles, multilook numbers, and subaperture overlap angles. Using this method, the Vehicle-Det dataset collected by our team and the publicly available SAR parking lot dataset Gotcha [37] are reimaged from the original echo. Then, the omnidirectional vehicle target detection dataset SAR-Vehicle-Det is constructed to evaluate algorithms effectively. This article also proposes an ASC-based guided network model ASC-U^{2}Det for SAR target detection, where the network prediction is divided into the ASC reconstruction branch and the target detection branch. By extracting ASC features of vehicle targets, ASC target reconstruction mask maps for large scenes are produced, and these ASC mask maps serve as supervised information to guide the network in extracting SAR target scattering features, suppressing background clutter interference, and improving localization accuracy. Meanwhile, the network also has certain physical interpretability. The main contributions of this article can be summarized as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

1) This article designs an SAR subaperture division data enhancement method with different aspect accumulation angles, multilook numbers, and subaperture overlap angles based on the original echo, and this method can obtain the omnidirectional subaperture images of targets within the observation scene, providing a novel solution for the expansion and construction of SAR datasets.
2) A novel network structure, ASC-U^{2}Det, is proposed in this article, which uses ASC physical knowledge as supervisory information to guide the network in extracting SAR target scattering features at different aspect angles, suppressing background clutter, and improving localization accuracy.

The rest of this article is organized as follows. Section II introduces the SAR subaperture division data enhancement method and the construction process of the SAR-Vehicle-Det dataset. Section III details the ASC extraction algorithm, the proposed model structure, and the training details. Section IV is the experimental part, where comparative and ablation experiments are designed to validate the model detection performance on the SAR-Vehicle-Det and miniSAR datasets. Finally, Section V concludes the article.

II. SAR Subaperture Division and Dataset Construction

In this section, the SAR omnidirectional subaperture division data enhancement method is first introduced, followed by the basic information of the Vehicle-Det dataset collected by our team, the publicly available SAR dataset Gotcha, and the specific details of constructing the SAR-Vehicle-Det target detection dataset.

A. SAR Omnidirectional Subaperture Division Data Enhancement

After obtaining the omnidirectional echoes of the target scene through SAR multiview observations, it is generally necessary to divide the subaperture before imaging. The subaperture division is related to the aspect accumulation angle, multilook number, and the overlap angle between adjacent subapertures. Fig. 3 illustrates the proposed subaperture division method, where the 360° aspect is divided into multiple subapertures, with each subaperture corresponding to an SAR image. Different combinations of multilook numbers (look), aspect accumulation angles (angle), and overlap angles (overlap) can obtain different subaperture images. Let the number of omnidirectional SAR subaperture divisions be denoted as N_{sub}, and the calculation equation is given as follows:

$$N_{\text{sub}} = \left\lfloor \frac{360^\circ}{\text{Angle} \times \text{Look} - \text{Overlap}} \right\rfloor$$
$$\text{Overlap} \in \left[0, \text{Angle} \times \text{Look}\right). \tag{1}$$

The larger the overlap angle in the equation, the more SAR images can be obtained. Furthermore, when the SAR system performs omnidirectional observations of the same area at different pitch angles, the subaperture division can be carried out for the echo at each pitch angle, resulting in an increased number of SAR images obtained. Assuming that the number of pitch angles is N_φ and the subaperture division method is the same for each pitch angle, then the final number of SAR images obtained is given by the following equation:

$$N_{\text{total}} = N_\varphi \times N_{\text{sub}}. \tag{2}$$

This article will use the subaperture division data enhancement method to obtain a large number of SAR images to support network training and evaluation.

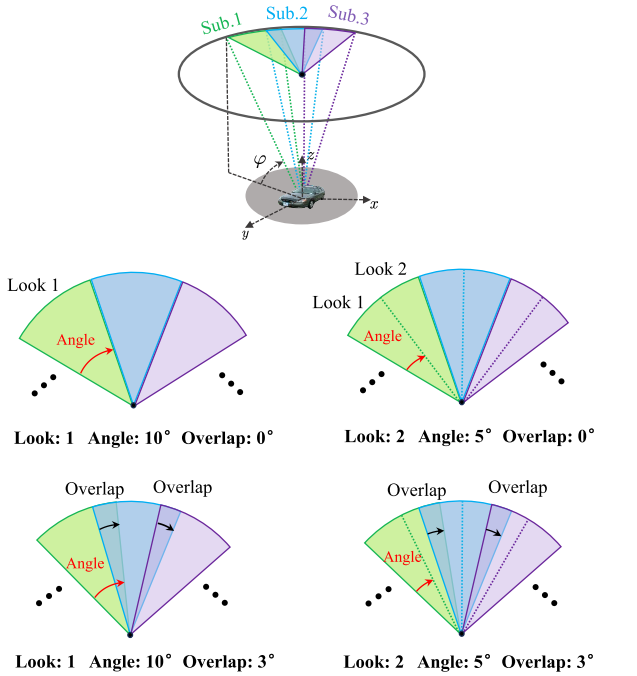


Fig. 3. Diagrams of the omnidirectional subaperture division, where different colors represent different subapertures. A 3-D illustration of the divided subaperture is provided at the top, while 2-D illustrations of four division methods are listed below. (Look: the number of multilook; angle: the aspect accumulation angles; and overlap: the overlap angles between adjacent subapertures.)

In terms of imaging methods, this article adopts the BP algorithm [38] for imaging of subaperture images. From the perspective of subaperture accumulation, SAR 2-D imaging modes are mainly divided into coherent processing and incoherent processing [39], [40]. Coherent processing performs coherent accumulation on apertures with a certain accumulation angle. Incoherent processing divides into multiple subapertures and images them separately, and the obtained subimages are noncoherently accumulated to get the final image. Incoherent accumulation can be regarded as a way of multilook processing, resulting in smoother SAR images with less coherent speckle noise [41]. Therefore, this article adopts coherent processing for single-look subaperture images and incoherent accumulation for multilook subaperture images.

B. Construction of the SAR-Vehicle-Det Dataset

The SAR-Vehicle-Det dataset constructed in this article consists of vehicle data collected by our team and publicly available Gotcha parking lot data, encompassing three scenes with approximately 200 vehicle targets. The SAR images and optical images of some scenarios from the SAR-Vehicle-Det dataset are shown in Fig. 4. The radar system used by our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

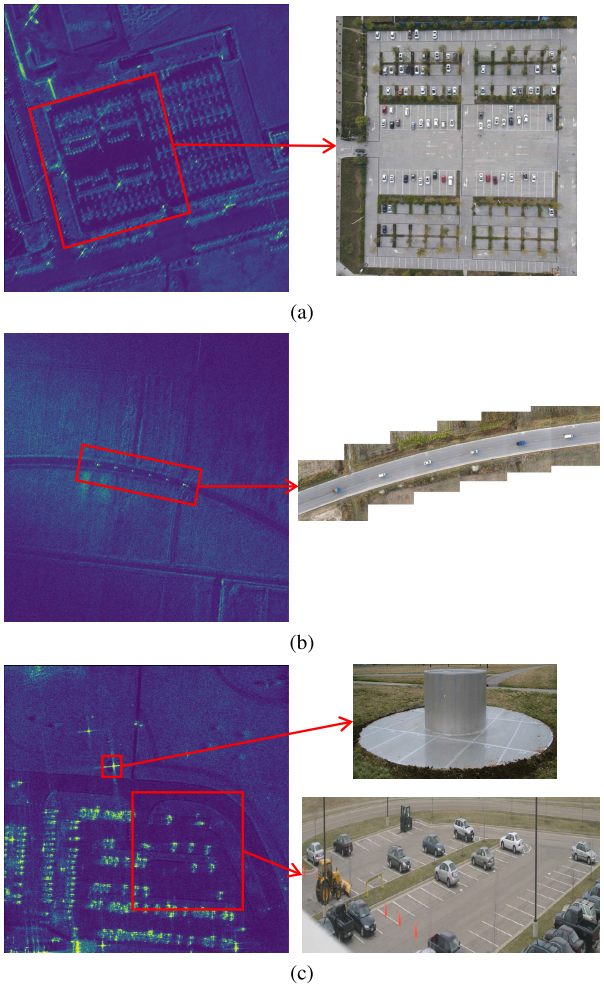


Fig. 4. SAR images and optical images of the observed scene from the SAR-Vehicle-Det dataset. (a) and (b) Images of urban car parks and highway scenes collected by our team. (c) Images of the publicly available Gotcha dataset.

team for collecting SAR data operates in the Ku-band, with a circular flight trajectory, and the observation scenarios include urban parking lots and highways, as shown in Fig. 4(a) and (b), where the observed targets are all civilian vehicles, including types such as hatchbacks and sedans. The Gotcha [37] data are publicly available full-polarization X-band SAR data collected by the Air Force Research Laboratory (AFRL), as shown in Fig. 4(c). The center frequency of the radar system is 9.6 GHz, and the bandwidth is 640 MHz. The experiments involve eight circular trajectories at different altitudes, with pitch angles ranging from 43.7° to 45°.

In this article, the echo data are reimaged using the subaperture division method, as described in Section II-A. Fig. 5 shows the SAR images of vehicle targets with different multilook numbers and aspect accumulation angles. The multilook incoherent accumulation reduces the coherent speckle noise in the images, but the scattering features are smoothed, resulting in slightly blurred images compared to single-look images. When the subaperture accumulation angle is small, the SAR image resolution is poor, and the image is greatly affected by coherent speckle noise. With the increase in the accumulation angle, the target scattering features gradually become more obvious. However, when the accumulation angle

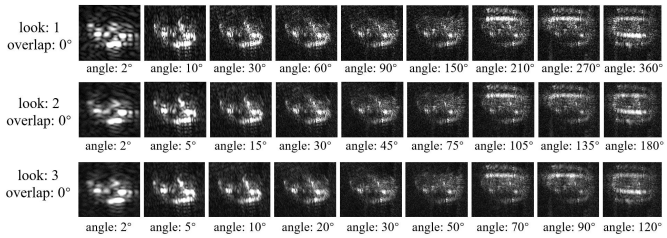


Fig. 5. SAR images of vehicle targets with different multilook numbers and aspect accumulation angles.

TABLE I
NUMBER OF SAR IMAGES FOR DIFFERENT SUBAPERTURE IMAGING PARAMETERS

Sub-aperture	Num	Sub-aperture	Num
1look-2°angle-0°olp	180	1look-10°angle-2°olp	45
1look-4°angle-0°olp	90	1look-10°angle-4°olp	60
1look-6°angle-0°olp	60	1look-10°angle-6°olp	90
1look-8°angle-0°olp	45	1look-10°angle-8°olp	180
1look-10°angle-0°olp	36	2look-10°angle-0°olp	18
1look-12°angle-0°olp	30	3look-10°angle-0°olp	12
1look-14°angle-0°olp	25	4look-10°angle-0°olp	9
1look-16°angle-0°olp	22	5look-10°angle-0°olp	7
1look-20°angle-0°olp	18	6look-10°angle-0°olp	6
1look-25°angle-0°olp	14	8look-10°angle-0°olp	4
1look-30°angle-0°olp	12	10look-10°angle-0°olp	3
1look-35°angle-0°olp	10	12look-10°angle-0°olp	3
1look-40°angle-0°olp	9	14look-10°angle-0°olp	2

becomes large, the scattering features of the targets in different aspect angles are more easily averaged. Therefore, this article primarily considered the resolution factor in the construction of the dataset. The resolution of the SAR range (R_x) and aspect (R_y) directions is calculated in the following equation:

$$R_x = \frac{C}{2B}$$
$$R_y = \frac{C}{4 f_c \sin(\varphi_m/2)} \tag{3}$$

where C represents the speed of light, B denotes the radar bandwidth, f_c is the center frequency, and φ_m is the aspect accumulation angle. To ensure high resolution in both range and aspect directions, which facilitates the feature extraction for subsequent detection and recognition tasks, SAR images with a single look of 10° aspect accumulation angle are used as the baseline data in this article. Calculations reveal that the aspect resolution has achieved 0.1 m, ensuring that the local scattering information of the target is essentially preserved.

The constructed SAR-Vehicle-Det contains civilian vehicles, and the target locations and category annotation format are consistent with the PASCAL VOC dataset format [42]. All SAR images are uniformly cropped to 800 × 800 pixels. To evaluate the impact of different subaperture accumulation angles and multilook imaging parameters on the target detection model, the SAR-Vehicle-Det dataset contains a variety of imaging results. Table I summarizes the subaperture division and the corresponding number of SAR images under a single

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

flight trajectory, with the number of SAR images calculated using (1), where the "1look-10°angle-2°olp" represents a single-look imaging method with 10° subaperture accumulation angle and 2° overlap angle, and the rest are analogous. Specific experimental verification and analysis are provided in Section IV.

III. Method of ASC-U^{2}Det

In this section, first, the SAR ASC model and the extraction method of the ASC reconstruction mask map for vehicle targets in large scenes are introduced. Second, a novel network ASC-U^{2}Det based on ASC physical guidance is proposed for vehicle target detection. Finally, the loss function and training details of the proposed network are described.

A. ASC Model

According to the ASC model theory [43], the target's backscattering response in the high-frequency conditions can be approximated as the sum of responses from multiple individual scattering centers. In the ASC model, the backscattering response is a function of frequency and aspect angle, which is represented as follows:

$$E(f,\varphi;\Theta)=\sum_{i=1}^{q}E_i(f,\varphi;\theta_i) \tag{4}$$

$$\begin{aligned}E_i(f,\varphi;\theta_i)=A_i\cdot&\left(j\frac{f}{f_c}\right)^{\alpha_i}\\&\cdot\exp\left[-j\frac{4\pi f}{C}(x_i\cos\varphi+y_i\sin\varphi)\right]\\&\cdot\operatorname{sinc}\left(\frac{2\pi f}{C}L_i\sin(\varphi-\bar{\varphi}_i)\right)\\&\cdot\exp(-2\pi f\gamma_i\sin\varphi)\end{aligned} \tag{5}$$

where q denotes the number of scattering centers. f and φ denote the radar frequency and the aspect angle, respectively. f_c is the radar center frequency, and C is the light speed. $E(f,\varphi;\Theta)$ denotes the backscattering response of the target. Θ is the set of all parameters, and $[A_i,\alpha_i,x_i,y_i,L_i,\varphi_i,\gamma_i]$ constitutes the parameter set of the ith scattering center, where A_i is the amplitude of the scattering center and α_i reflects the frequency dependence. (x_i,y_i) are the distance and azimuthal location coordinates. L_i and $\bar{\varphi}_i$ denote the length and angle of the distributed scattering center, respectively. γ_i reflects the aspect dependence.

The joint optimization of the scattering center parameters is a nonlinear nonconvex parameter estimation problem [44]. Existing ASC extraction methods are mainly divided into image-domain methods and frequency-domain methods [45], [46]. Considering the extraction speed, this article adopts the approximate maximum likelihood (AML) algorithm [47] to extract ASC for vehicle targets, and the specific extraction process is summarized in Algorithm 1.

B. Extraction of ASC Reconstruction Mask Maps

Mask maps are commonly used as outputs of semantic segmentation networks [34], and the mask assigns a semantic label to each pixel, which helps the network to understand targets finely and accurately. Inspired by mask, this article constructs the ASC reconstruction mask maps and uses them as supervisory information to guide the network in extracting SAR target scattering features. Semantic segmentation mask maps typically require manual annotation of target contours, which can be time-consuming. For SAR images, the mask map only annotated with the target contour cannot describe the target scattering features. Therefore, this article proposes a method for automatically extracting ASC reconstruction mask maps.

Algorithm 1 ASC Parameter Extraction Algorithm

Input: SAR complex data $I(x,y)$, scattering center number k, reconstruction residual threshold t.
Output: ASC model parameter set Θ.
Initialization: scattering center number $i=1$, reconstruction residual image $d_{res}=|I(x,y)|$, ASC model parameter set Θ.
While $i<k$ & $\|d_{res}\|\geq t$
1. Calculate SAR data dB values: $I_{dB}=20\cdot\log(|I|/\max(|I|))$
2. Split the region $d_{\max}(x,y)$ with the highest amplitude from the target region $I_{dB}(x,y)$ using the watershed algorithm.
3. Estimate the ASC model parameters θ_i using the AML algorithm.
4. Calculate the reconstruction images from the ASC parameters:$e_i(x,y;\theta_i)=IFFT_{2D}[E_i(f,\varphi;\theta_i)]$.
5. Update the reconstruction residual image: $d_{res}=d_{res}-e_i$.
6. Add the parameter θ_i to the set Θ, $i=i+1$.

EndWhile

Fig. 6 illustrates the extraction process of ASC reconstruction mask maps. First, the original SAR image is cropped into target slices according to the target annotations. Then, Algorithm 1 is used to extract ASC parameters from the target slice, and the ASC reconstructed image slices are obtained from the following equation:

$$d(m,n)=\mathrm{IFFT}_{2\text{-D}}\left[E(f,\varphi;\Theta)\right] \tag{6}$$

where (m,n) denotes the image-domain pixel coordinates and $\mathrm{IFFT}_{2\text{-D}}[\cdot]$ denotes the 2-D inverse Fourier transform. To facilitate subsequent use, the pixel values of the ASC reconstructed slices are uniformly normalized to [0, 1]. Finally, by combining the target annotations, all ASC reconstructed image slices are stitched on a pure black image to obtain a large-scale scene mask. The ASC reconstruction mask map only contains the scattering features of the target region, while the background clutters are completely suppressed.

C. Model Structure of ASC-U^{2}Det

Most convolutional neural networks (CNNs) typically reduce the resolution of feature maps to increase depth [48]. However, for SAR image vehicle targets, which are small in large scenes, the low resolution of the feature maps after

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

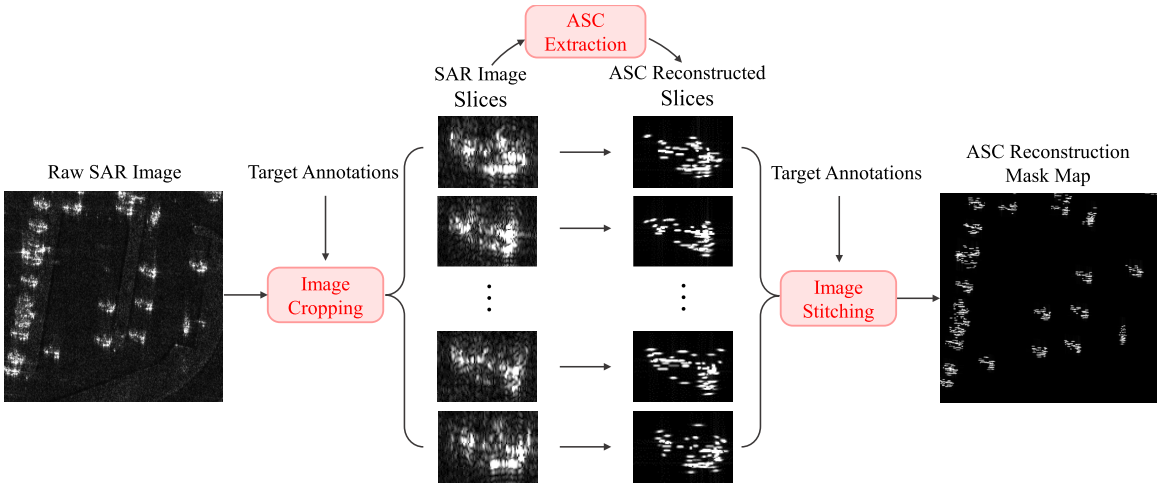


Fig. 6. Extraction process of the ASC reconstruction mask map.

passing through the deep network can affect feature extraction and target detection. In addition, to obtain ASC reconstruction masks of the same size as the input images, it is essential to maintain a high resolution of intermediate feature maps in the network. Therefore, inspired by U^2-Net [49], this article adopts a two-level nested U-shaped structure, which can extract multiscale features of the target without reducing the resolution of the feature maps.

Fig. 7 illustrates the ASC-U^{2}Det structure designed in this article, which consists of an encoder, a decoder, and prediction heads. The encoder and decoder correspond to the backbone network and reconstruction network, respectively. The network predictions are divided into the ASC reconstruction head and the target detection head. The ASC physical guidance method proposed in this article is a form of multitask learning [50]. Multitask learning is a paradigm in machine learning that aims to jointly learn multiple related tasks so that the knowledge contained within one task can be leveraged by others, thereby enhancing the generalization performance of all tasks. The ASC-U^{2}Det network involves two tasks: ASC mask map reconstruction and SAR target detection. Both task branches are conducted simultaneously during network training and share feature extraction parameters. Because the ASC reconstruction branch focuses on suppressing background clutter and extracting target scattering features, it influences the feature extraction network to pay more attention to target scattering information, which, in turn, affects the target detection branch and improves its detection performance. The following describes the structure of each component in detail.

1) RSU-N and RSU-4F Modules: Both modules are residual U-shaped structures, where N represents the depth of the U-shaped structure. The larger the value of N, the richer the local and global information extracted by the network. RSU-N can also be seen as composed of a small encoder and decoder. The encoder obtains feature maps through convolution modules and max-pooling [51] downsampling, and the decoder gradually obtains high-resolution multiscale feature maps through feature fusion, bilinear interpolation upsampling, and convolution modules. It is worth noting that the RSU-N uses dilated convolution [52] in the last layer of the encoder to expand the receptive field, which is beneficial for extracting global features. Compared to RSU-N, RSU-4F lacks pooling and upsampling operations and replaces several convolutional modules with dilated convolution to expand the receptive field.

2) Encoders and Decoders: The encoder and decoder of the ASC-U^{2}Det network are stacked by RSU-N and RSU-4F units, forming a large U-shaped structure. This nested U-shaped structure can effectively extract multiscale features and aggregate multilevel features between modules [49]. After multiple layers of encoder convolution and pooling, the resolution of the feature map decreases. To maintain the resolution of the feature map and expand the receptive field, the last two modules of the encoders adopt the RSU-4F structure. Each module of the decoders fuses the upsampled feature map from the previous stage with the output feature map from its symmetric encoder module. The decoders gradually obtain high-resolution feature maps through a progressive upsampling strategy, thereby alleviating the local feature loss caused by direct large-scale upsampling.

3) Prediction Heads: The prediction heads of the ASC-U^{2}Det network consist of the ASC reconstruction head and the target detection head. The ASC reconstruction head integrates the outputs of different modules in the decoders, as shown in Fig. 7. After each decoder module, the different resolution feature maps are recovered into the ASC feature maps of the same size as the raw SAR image through convolutional and upsampling layers. This process can be represented by the following equation:

$$m_i = \text{Upsample}(\text{Conv}(f_i)), \quad i = 1, 2, \ldots, 6 \tag{7}$$

where f_i and m_i denote the ith feature maps of the decoders and the ith ASC feature maps of the prediction heads, respectively. Upsample$(\cdot)$ represents bilinear interpolation upsampling, and Conv$(\cdot)$ represents the convolution operation. After concatenating the ASC feature maps from different stages of the prediction heads, they are fused through a convolutional layer to obtain the final ASC reconstruction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

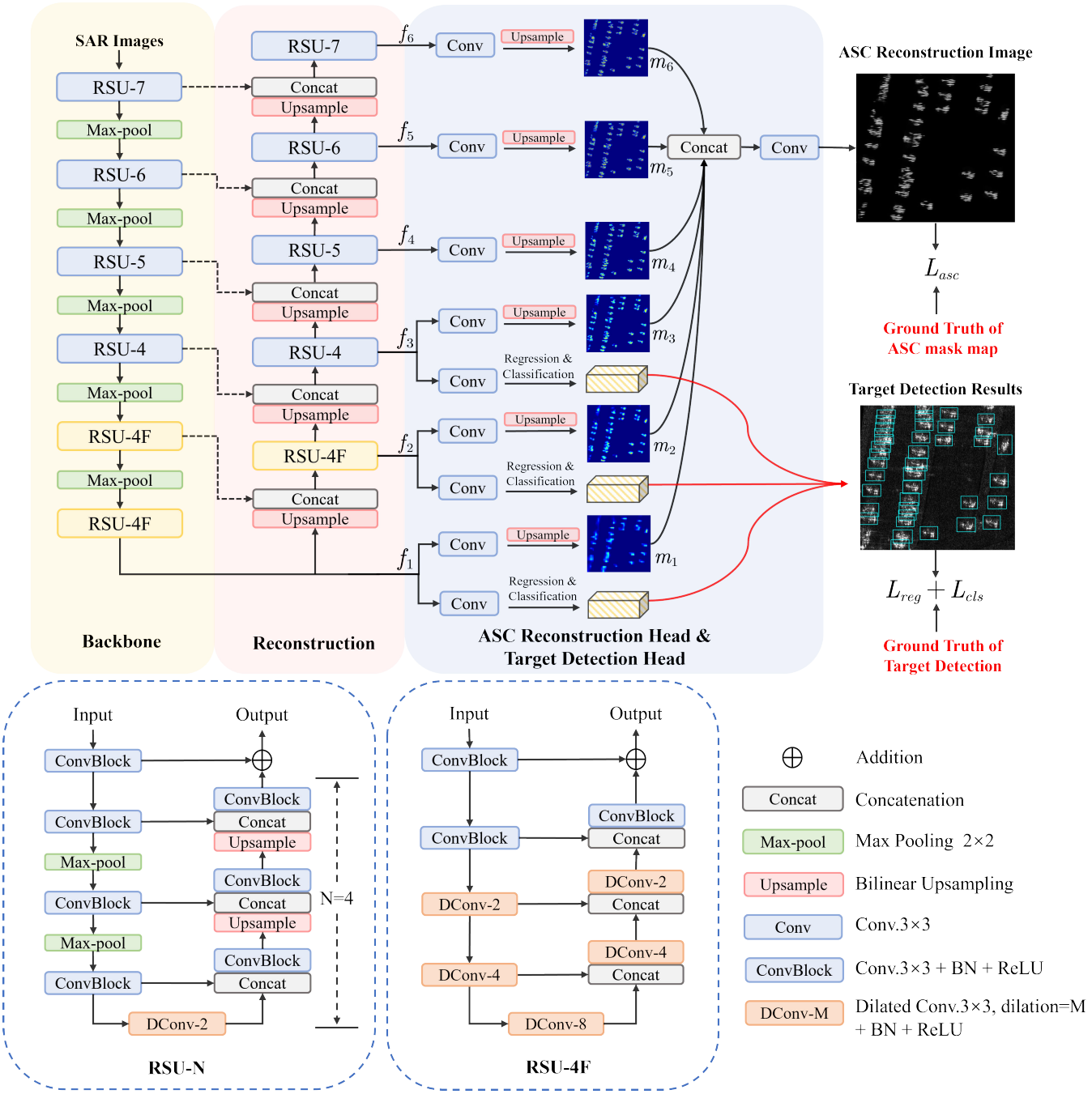


Fig. 7. ASC-U^{2}Det model architecture diagram. The overall framework includes the backbone network, the reconstruction network, the ASC reconstruction prediction head, and the target detection head. The RSU-N and RSU-4F represent the basic residual U-shaped modules. The different layers of the network structure are listed in the bottom right corner, where BN represents batch normalization and ReLU represents the activation function. L_{asc}, L_{reg}, and L_{cls} denote the ASC reconstruction loss, the target location regression loss, and the target classification loss, respectively.

image, as given by the following equation:

$$m_{\text{fuse}} = \text{Sigmoid}(\text{Conv}([m_1, m_2, \ldots, m_6])) \tag{8}$$

where m_{fuse} denotes the ASC reconstruction image. Sigmoid$(\cdot)$ denotes the activation function, which aims to normalize the ASC reconstruction image to the range of [0, 1]. For the target detection head, in order to adapt to targets of different sizes, this article performs classification and bounding box regression on three different resolution feature maps from the decoders.

D. Loss Function of ASC-U^{2}Det

Since the ASC-U^{2}Det has two prediction heads, the ASC reconstruction head and the object detection head, the overall loss function also contains multiple losses, which is similar to multitask learning [50]. The ASC reconstruction mask can be considered a priori knowledge with physical significance, which improves the accuracy of the target detection task through supervised constraints. The loss function of the model is defined as follows:

$$L = L_{\text{asc}} + L_{\text{reg}} + L_{\text{cls}} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

where L_{asc}, L_{reg}, and L_{cls} denote the ASC reconstruction loss, target location regression loss, and target classification loss, respectively. When training the neural network, the optimization objective is to minimize the total loss function of the model. The ASC reconstruction loss not only computes the loss of the network output ASC reconstruction image but also includes the losses of the ASC reconstruction maps at different stages of the decoders, as shown in the following equations:

$$L_{\text{asc}} = \sum_{i=1}^{M} \text{mse}(P_i, G) + \text{mse}(m_{\text{fuse}}, G) \tag{10}$$

$$\text{mse}(P, G) = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} \|P(x, y) - G(x, y)\| \tag{11}$$

$$P_i = \text{sigmoid}(m_i) \tag{12}$$

where P_i is the result of the ith ASC feature map of the decoders after the sigmoid function. M denotes the number of ASC feature maps generated by the decoders. H and W represent the width and height of the ASC reconstruction mask map, and G is the ground truth of the ASC reconstruction mask map. mse$(\cdot)$ denotes the mean square error loss function.

The target location regression loss is calculated using the CIoU_Loss [53] function as follows:

$$L_{\text{reg}} = \sum_{i}^{N_l} \text{CIoU}(P_{\text{box}}, T_{\text{box}}) \tag{13}$$

where $P_{\text{box}} \in \mathbb{R}^{N\times(x_c,y_c,w,h)}$ and $T_{\text{box}} \in \mathbb{R}^{N\times(x_c,y_c,w,h)}$ denote the bounding boxes predicted by model and the corresponding bounding boxes of ground truth, respectively. N represents the number of targets. N_l represents the number of prediction layers for target detection. The classification loss is calculated with binary cross-entropy (BCE) loss as follows:

$$L_{\text{cls}} = \sum_{i}^{N_l} \text{BCE}(P_{\text{cls}}, T_{\text{cls}}) \tag{14}$$

$$\text{BCE}(P_{\text{cls}}, T_{\text{cls}}) = -\frac{1}{N} \sum_{j=1}^{N} \left[p_j \log\left(t_j\right) + \left(1 - p_j\right) \log\left(1 - t_j\right)\right] \tag{15}$$

where $P_{\text{cls}} \in \mathbb{R}^{N\times N_c}$ and $T_{\text{cls}} \in \mathbb{R}^{N\times N_c}$ denote the prediction and ground-truth category labels, respectively. N_c is the number of vehicle target categories. p_j and t_j denote the predicted and true category vector. Although the proposed ASC-U^{2}Det has two prediction heads, the training parameters are trained in the same way as a normal network, and backpropagation [54] can still be used to compute gradients and update the network parameters.

IV. Experiments and Analysis

In this section, the performance of the proposed method is evaluated using the SAR-Vehicle-Det dataset constructed in Section II and the public dataset miniSAR. First, the experimental setup and evaluation metrics are introduced. Then, the effectiveness of the proposed model is verified through comparative experiments and ablation experiments. Finally, the impact of different subaperture accumulation angles and multilook imaging parameters on the network detection performance is evaluated.

TABLE II
Number of Train and Test Sets on Four Experiments

EXP	Train		Test	
	Sub-aperture	Num	Sub-aperture	Num
EXP 1	1look-10°angle-0°olp	310	1look-10°angle-0°olp	78
EXP 2	1look-10°angle-0°olp	310	1look-10°angle-8°olp	386
	1look-10°angle-2°olp	383		
	1look-10°angle-4°olp	510		
	1look-10°angle-6°olp	771		
	1look-10°angle-8°olp	1535		
EXP 3	1look-10°angle-8°olp	1535	1look-2°angle-0°olp	356
			1look-4°angle-0°olp	356
			1look-6°angle-0°olp	356
			…	…
			1look-35°angle-0°olp	356
			1look-40°angle-0°olp	356
EXP 4	1look-10°angle-8°olp	1535	2look-10°angle-0°olp	178
			4look-10°angle-0°olp	178
			6look-10°angle-0°olp	178
			…	…
			12look-10°angle-0°olp	178
			14look-10°angle-0°olp	178

A. Experimental Setup

The constructed SAR-Vehicle-Det dataset includes SAR images with different subaperture angles, multilook numbers, and overlap angles. The image size is 800 × 800 pixels, and the labels include target detection annotations and ASC reconstruction mask images. Table II lists a summary of the experimental setup dataset. The four experiments designed in this article are given as follows.

1) EXP 1 (Baseline Experiment): In this article, SAR images with single-look 10° subaperture accumulation angle and nonoverlapping between adjacent subapertures (1look-10angle-0olp) are used as the baseline data. These images are divided into the train and test sets in an 8:2 ratio. Subsequent experiments will be compared with this baseline data.

2) EXP 2 (Different Overlap Angles): The dataset is expanded by overlapping adjacent subapertures while ensuring that each subaperture image is different. In this experiment, the number of multilook and the subaperture accumulation angle are kept constant, and only the overlap angles are different. The main purpose is to verify the impact of overlap angles on the performance of networks.

3) EXP 3 (Different Subaperture Accumulation Angles): The subaperture accumulation angle affects the resolution of SAR images. Smaller accumulation angles result in stronger coherent speckle noise in the images, while larger accumulation angles make it easier to average the scattering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

characteristics of different aspect angles. In this experiment, the training set remains consistent, while the test set consists of SAR images with different subaperture accumulation angles. The main focus is to explore the impact of subaperture accumulation angles on the performance of networks.

4) EXP 4 (Different Multilook Numbers): Multilook processing is commonly used to suppress SAR coherent speckle noise, and the processed SAR images are smooth and slightly blurred compared to single-look images. In this experiment, the training set remains consistent, while the test set consists of SAR images with different numbers of multilook. The aim is to investigate the impact of multilook imaging on the performance of networks.

To eliminate the bias introduced by the different number of test sets, this article randomly cropped and sampled different subaperture SAR images for EXP 3 and EXP 4 to maintain consistency in the number of test images. Specifically, when the number of test images is small, the cropping keeps the larger overlapping pixels, and when the number of test images is large, the random sampling method is used after cropping to ensure that the number of test images under different subaperture divisions is the same.

In these experiments, several commonly used target detection algorithms are selected to compare with the proposed ASC-U^{2}Det, including the two-stage algorithm Faster-RCNN [55], the one-stage algorithm YOLOv5 of the YOLO series [56], the anchor-free algorithm FCOS [57], the YOLOv8 of the YOLO series [56], and the transformer-based algorithm DINO [58]. In addition, this article also selected recently released SAR detectors with better performance for comparison, namely, DiffDet4SAR [59], MSFA [60], and DenoDet [61]. The MSFA model chooses to use the Faster-RCNN-based structure.

In this article, all experiments were implemented in PyTorch on the same computer, which consists of an Intel Core i9-13900K CPU with 3-GHz frequency and an Nvidia GeForce RTX 4090 GPU with 24.0-GB memory. To objectively evaluate the common model performance, all common models did not use pretrained weights. During the training phase, the parameter settings in this article were consistent with the YOLO model [56] using the initial learning rate of 1×10^{-2}, a weight decay of 5×10^{-4}, and a momentum of 0.937 to update model weights. For the particular SAR detectors, parameters such as the learning rate were consistent with those in their article.

B. Evaluation Metrics

This article adopts the precision, recall, F1 score, and average precision (AP) to evaluate the target detection performance of different methods. The definitions of precision and recall are given as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

where true positives (TPs), false positives (FPs), and false negatives (FNs) represent the number of correctly detected targets, false alarms, and missed targets. The F1 score considers both precision and recall, defined as follows:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{18}$$

The AP is a commonly used metric to evaluate the comprehensive performance of the model and is defined as follows:

$$\text{AP} = \int_0^1 P(R) dR \tag{19}$$

where P represents precision, R represents recall, and AP is defined as the region under the precision–recall curve.

The proposed ASC-U^{2}Det not only has target detection output but also has ASC reconstruction image output. Therefore, this article uses mean absolute error (MAE) [62] to evaluate the quality of the ASC reconstruction images output by the network. The MAE is defined as follows:

$$\text{MAE}(P, G) = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} |P(x, y) - G(x, y)| \tag{20}$$

where P and G represent the network-predicted ASC reconstruction image and the corresponding ground truth, H and W are the height and width of the image, and (x, y) represents the pixel coordinates. A smaller MAE indicates a higher quality of reconstructed ASC images by the model.

To evaluate the impact of different imaging methods on SAR image quality, this article also uses the peak signal-to-noise ratio (PSNR), the structural similarity measure index (SSIM) [63], and AE [10] as evaluation metrics of image similarity. The PSNR is defined as follows:

$$\text{PSNR}(X, Y) = 20 \cdot \log_{10}\left(\frac{X_{\max}}{\sqrt{\text{mse}(X, Y)}}\right) \tag{21}$$

where X and Y represent the two images to be compared, $X_{\max}$ represents the maximum pixel value of the image, and the calculation of mse is shown in (11).

The AE is used to measure the anisotropic scattering of SAR subaperture images, defined as follows:

$$\text{AE} = \text{mean}\left(-\sum_{k=1}^{n} p(k) \log_n p(k)\right) \tag{22}$$

$$p(k) = \frac{X(k)}{\sum_{k=1}^{n} X(k)} \tag{23}$$

where mean$(\cdot)$ denotes the averaging operation, $p(k)$ denotes the amplitude probability of the image, X represents the subaperture image, and n is the number of subaperture images. A higher value of three evaluation metrics indicates higher similarity between the images.

C. Baseline Experiment (EXP 1) Results

Table III shows the comparative experimental results of the proposed ASC-U^{2}Det model with other advanced target detection networks. It can be seen that the proposed method outperforms other target detection methods in all evaluation metrics. The performance of Faster-RCNN is the lowest because it belongs to the two-stage network with a large number of training parameters. In addition, no pretrained weights

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III
DETECTION RESULTS OF DIFFERENT METHODS UNDER EXP 1

Method	P	R	F1	AP	Params(M)
Faster-RCNN [55]	56.97	57.67	57.32	47.30	41.35
FCOS [57]	76.06	72.26	74.11	75.60	32.11
YOLOv5 [56]	89.21	89.29	89.25	91.98	7.01
YOLOv8 [56]	92.60	82.09	87.03	89.69	11.13
DINO [58]	87.41	78.95	82.96	74.00	47.54
DiffDet4SAR [59]	89.87	91.26	90.56	89.87	110.92
MSFA(frcnn) [60]	83.66	86.45	85.03	87.50	41.60
DenoDet [61]	91.26	89.52	90.38	90.50	65.63
ASC-U^{2}Det	**91.01**	**90.63**	**90.81**	**95.03**	**19.45**

are used in the experiments, and the training set samples are relatively small, leading to poor model performance. Similarly, the transformer-based DINO model also performs poorly due to the excessive number of parameters. FCOS and YOLOv8 are anchor-free detection models, which lack predefined anchor boxes for the target, leading to lower detection rates and localization accuracy compared to YOLOv5. YOLOv5 is a one-stage detector with adaptive anchor box calculation, multiscale feature fusion modules, and a pyramid module that fuses high-level semantic information and low-level features, hence its relatively better performance. DiffDet4SAR, MSFA, and DenoDet are detectors designed for SAR images, which are improved in terms of enhancing the saliency of anisotropic targets, model transfer, and multisubspace features denoising in the frequency domain, respectively, and achieved good results compared to ordinary detectors. The proposed ASC-U^{2}Det guides the network to extract the physical scattering features of SAR targets in different aspect angles, and the ASC mask image contains natural positional information that can also help the network in precise localization. Even without data expansion using subaperture overlap, the proposed method still outperforms YOLOv5 by approximately 3% in evaluation indicators, demonstrating good detection performance.

D. Different Overlap Angle (EXP 2) Results

In the setup of EXP 2, the test set consists of omnidirectional subaperture SAR images with an overlap angle of 8°, aimed at simulating real-world scenarios where targets may be observed from any aspect angle. Subaperture overlap means that the starting points of the accumulation angles are different. Although adjacent subapertures have common areas, their scattering characteristics reflected in the image are not completely identical. Fig. 8 shows the similarity curves of adjacent subaperture images under three evaluation metrics, and the PSNR metric is normalized, where BMP represents SAR gray-scale images and ASC represents ASC reconstruction images. It can be observed that as the overlapping angle increases, the adjacent subaperture images become more similar, but differences still exist. This means that the scattering characteristics of subaperture images vary significantly under different aspect angles.

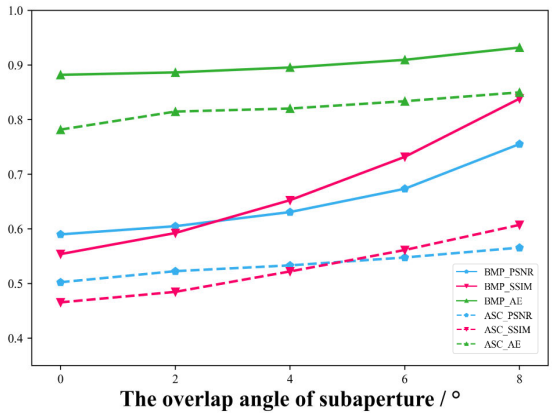


Fig. 8. Similarity curves of adjacent subaperture images at different overlap angles. (BMP: gray-scale SAR images; ASC: ASC reconstruction images.)

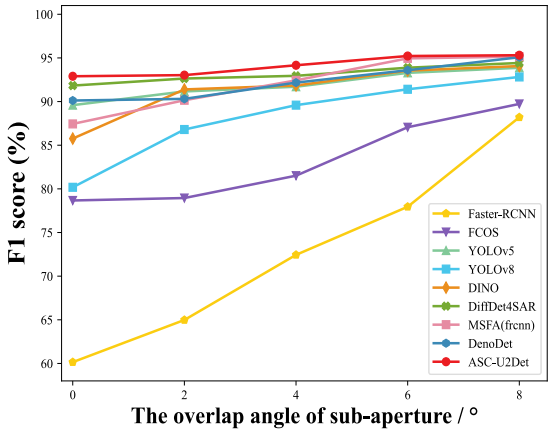


Fig. 9. F1 curves of each model for different overlap angle training sets.

Table IV and Fig. 9 demonstrate the detection performance of the models under different subaperture overlap angles in the training set. With the increase in the subaperture overlapping angle, the training dataset encompasses subaperture images of the target from all aspect angles, allowing the network to learn the omnidirectional scattering features of the target, thus improving the detection performance of all models. Especially for the models with a large number of parameters, Faster-RCNN, FCOS, YOLOv8, and DINO, the improvement is very significant, and the AP metrics increased by approximately 35%, 11% 18%, and 18%, respectively. In contrast, due to the good performance of YOLOv5, YOLOv8, DiffDet4SAR, DenoDet, and ASC-U^{2}Det, the evaluation metrics only show a small improvement. The proposed ASC-U^{2}Det can effectively extract ASC features of different aspect angles. As indicated by the MAE evaluation in Table V, with the expansion of the dataset, the output error of the network's ASC reconstruction branch is also reduced, and while the network detection performance improves, the ASC reconstruction capability also improves.

E. Different Subaperture Accumulation Angle (EXP 3) Results

The subaperture accumulation angle is closely related to the SAR image resolution. When the subaperture accumulation angle is small, the SAR image resolution is poor,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
DETECTION RESULTS OF MODELS WITH DIFFERENT SUBAPERTURE OVERLAP ANGLE TRAINING SETS

Method	Overlap angles	P	R	F1	AP
Faster-RCNN [55]	0°	59.48	60.84	60.15	52.50
	2°	66.86	63.20	64.98	58.80
	4°	73.46	71.44	72.44	71.30
	6°	77.43	78.47	77.95	79.90
	8°	84.99	86.63	85.80	88.20
FCOS [57]	0°	79.26	78.09	78.67	79.10
	2°	81.71	76.37	78.95	79.10
	4°	80.90	82.12	81.51	84.30
	6°	85.17	89.05	87.06	90.40
	8°	88.09	91.43	89.73	92.70
YOLOv5 [56]	0°	90.80	88.41	89.59	91.34
	2°	92.66	89.70	91.16	92.55
	4°	92.69	90.73	91.70	94.83
	6°	94.09	92.51	93.29	95.60
	8°	94.42	93.33	93.87	95.73
YOLOv8 [56]	0°	87.05	74.31	80.18	77.89
	2°	90.38	83.50	86.80	88.44
	4°	91.80	87.50	89.59	92.51
	6°	92.51	90.33	91.41	94.40
	8°	94.00	91.66	92.82	95.82
DINO [58]	0°	89.60	82.33	85.76	77.20
	2°	93.07	89.78	91.39	83.80
	4°	92.65	91.04	91.84	87.40
	6°	95.11	92.02	93.54	91.20
	8°	97.41	90.96	94.07	95.60
DiffDet4SAR [59]	0°	91.56	92.12	91.83	92.12
	2°	92.02	93.26	92.64	93.15
	4°	92.35	93.56	92.95	94.61
	6°	93.36	94.42	93.88	95.89
	8°	93.44	95.43	94.42	96.45
MSFA(frcnn) [60]	0°	87.27	87.60	87.44	88.10
	2°	90.80	89.50	90.15	90.20
	4°	91.94	92.93	92.43	93.30
	6°	94.04	95.88	94.95	95.90
	8°	93.15	97.23	95.15	97.10
DenoDet [61]	0°	89.06	91.21	90.12	92.10
	2°	89.21	91.44	90.31	92.30
	4°	90.87	93.52	92.17	93.90
	6°	92.65	94.57	93.59	95.30
	8°	93.18	97.05	95.08	97.30

TABLE V
DETECTION RESULTS OF ASC-U^{2}DET WITH DIFFERENT SUBAPERTURE OVERLAP ANGLE TRAINING SETS

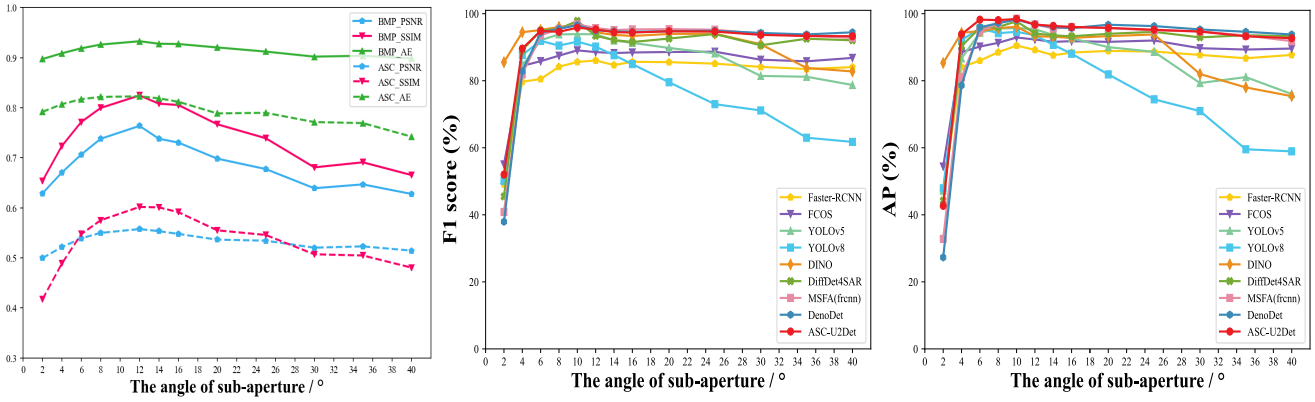
Method	Overlap angles	P	R	F1	AP	MAE↓
ASC-U^{2}Det	0°	91.79	94.03	92.90	95.65	0.0093
	2°	91.57	94.54	93.03	95.74	0.0091
	4°	93.15	95.19	94.16	97.11	0.0097
	6°	93.62	96.89	95.22	98.18	0.0086
	8°	93.22	97.48	95.31	98.37	0.0084

and when the accumulation angle is too large, the scattering characteristics of targets in different aspect angles are more easily averaged. In this article, SAR images with 10° subaperture accumulation angle are chosen as the reference images. Fig. 10(a) shows the similarity curves of images with different accumulation angles compared to the 10° accumulation angle images. It can be seen that for both BMP images and ASC reconstruction images, the similarity curves trend to increase first and then decrease, which is consistent with the previous analysis. Fig. 10(b) and (c) plots the F1 curves and AP curves of various models under different subaperture accumulation angles. The trends of evaluation metrics curves are consistent with the similarity curves. When the subaperture accumulation angle is small, the model performance sharply declines. Near the 10° accumulation angle, the model can maintain good generalization. As the accumulation angle continues to increase, the model performance gradually decreases. To ensure that the model performs well in practical applications, the subaperture accumulation angle for imaging should be close to the subaperture accumulation angle for model training.

F. Different Multilook Number (EXP 4) Results

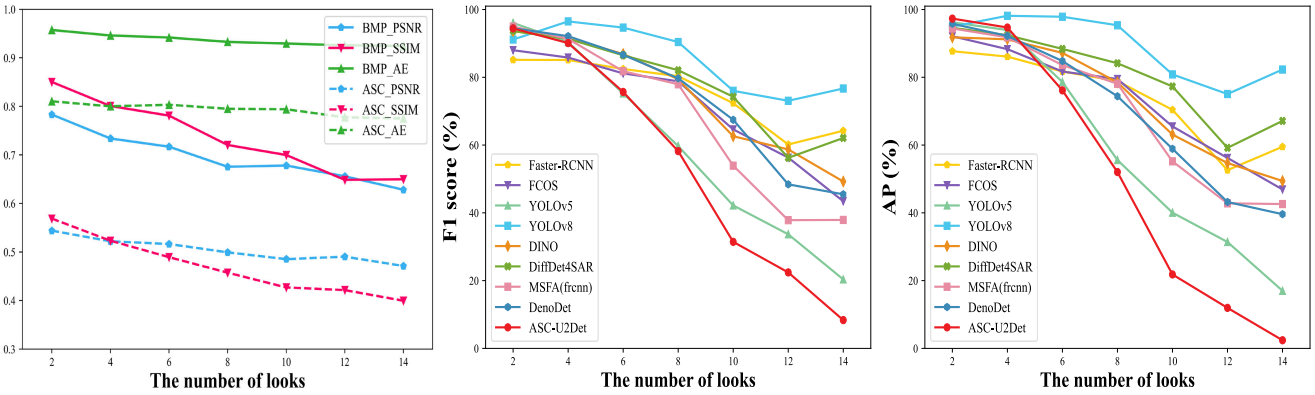
Multilook is commonly used to suppress speckle noise, and SAR images processed with multilook are smoother. Fig. 11(a) shows the similarity curve between multilook and single-look SAR images, and as the number of looks increases, the image difference becomes larger. Fig. 11(b) and (c) also plots the F1 curve and AP curve of each model under different numbers of looks. It can be observed that as the number of looks increases, all model evaluation metrics decrease to varying extents. Compared with the performance change curves of the subaperture accumulation angle in Fig. 10, multilook has a greater impact on model performance. The reason is that the target features are smoothed after multilook processing, and the distribution difference between the model training and test sets leads to a decrease in detection performance. Notably, the proposed ASC-U^{2}Net is more affected by multilook because the proposed model pays more attention to the scattering features of targets in the training set. The scattering features of targets are more pronounced in single-looking images, and the scattering features of targets in multilook images differ greatly from those in single-look images, which affects detection performance. However, in practical imaging, large multilook numbers are not usually selected. When the number of looks is small, the proposed ASC-U^{2}Det can still maintain good performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 10. (a) Similarity curves between accumulation angle images of different subapertures and the 10° accumulation angle images. (b) F1 curves of each model under different subaperture accumulation angles. (c) AP curves of each model under different subaperture accumulation angles.



(a) (b) (c)

Fig. 11. (a) Similarity curves between different multilook images and single-look images. (b) F1 curves of each model under different multilook images. (c) AP curves of each model under different multilook images.

G. Results on the miniSAR Dataset

To further validate the effectiveness of the proposed method, we supplemented experiments on the public dataset miniSAR [17]. The miniSAR dataset is a collection of actual measured data in the Ku-band spotlight mode. The resolutions of SAR images in the miniSAR are 0.1 × 0.1 m, encompassing a large area with complex scenes. The background clutter includes both man-made (such as buildings, roads, and streetlights) and natural clutter (such as trees and grass), which can well verify the model's performance in complex scenes. In terms of the experimental setup, the large original SAR images in the miniSAR are cropped to 512 × 512 pixels and divided into training sets and test sets, each with 576 and 144 images, respectively, while keeping the remaining experimental parameters consistent with Section IV-A. Since the miniSAR does not provide the raw echo data, the overlapping subaperture division is not used in this experiment.

Table VI shows the comparative experimental results of the proposed ASC-U^{2}Det model with other advanced target detection networks. The proposed method outperforms other detectors in terms of F1 and AP metrics. For complex scenes, Faster-RCNN and FCOS perform poorly with higher false alarm and miss rates. The DINO model based on transformers also underperforms due to excessive parameters. The DiffDet4SAR model designed a scattering feature enhancement module specifically for SAR images, outperforming general detectors in complex scenes. MSFA employs a multistage filtering enhancement pattern that effectively improves SAR target detection performance. DenoDet suppresses SAR coherent speckle noise in the feature space,

TABLE VI
DETECTION RESULT OF DIFFERENT METHODS ON THE MINISAR DATASET

Method	P	R	F1	AP
Faster-RCNN [55]	93.57	90.89	92.21	93.40
FCOS [57]	83.09	78.69	80.83	84.40
YOLOv5 [56]	94.80	93.92	94.36	96.45
YOLOv8 [56]	95.05	91.54	93.26	97.45
DINO [58]	90.30	87.93	89.09	89.60
DiffDet4SAR [59]	96.32	93.13	94.70	95.87
MSFA(frcnn) [60]	94.01	94.70	94.35	96.60
DenoDet [61]	97.06	93.04	95.01	97.50
ASC-U^{2}Det	**97.83**	**93.57**	**95.65**	**98.87**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

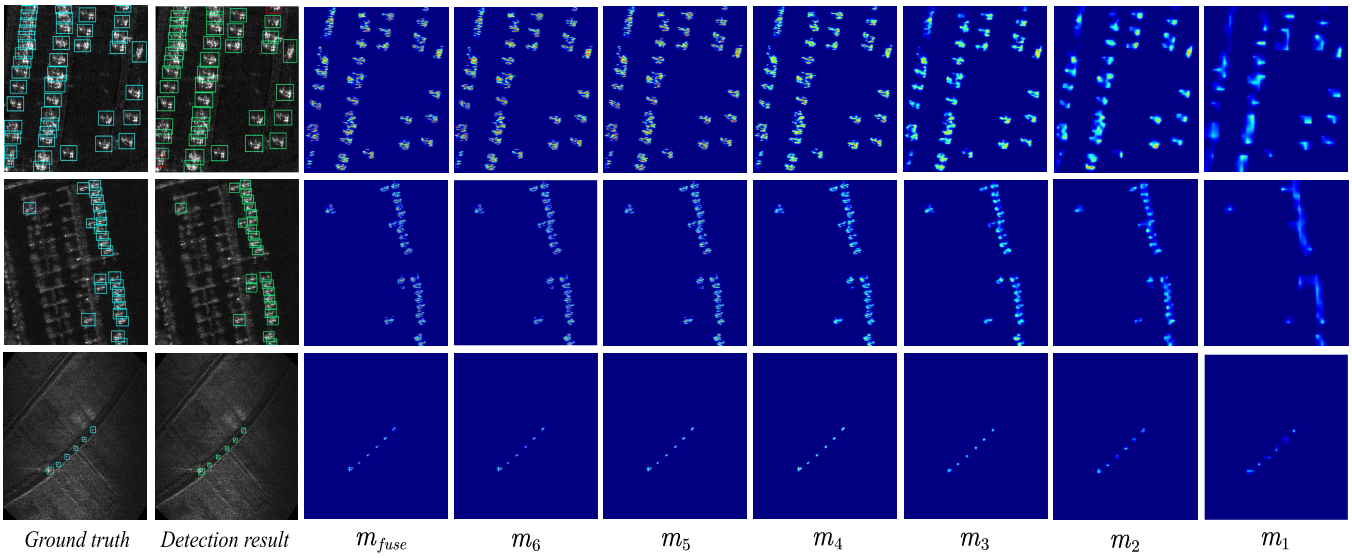


Fig. 12. Visualization results of the ASC-U^{2}Det output. Blue boxes represent ground truth, green boxes indicate detection results, and red boxes indicate the missed target. m_{fuse}: the output ASC reconstruction image; $m_1 \sim m_6$: feature maps at different stages of the decoders.

resulting in a significant enhancement in detection performance. The proposed ASC-U^{2}Det model directly utilizes ASC mask maps to guide the network to focus on target scattering features, suppressing background noise, and still outperforms DenoDet by approximately 1.37% in the AP metric, achieving the best performance among all compared methods. It is evident that the proposed method has some adaptability to complex scenes.

H. Visualization of ASC-U^{2}Det Output

To visually demonstrate the effectiveness of the proposed model, Fig. 12 shows the SAR image target detection results and the visualization of the model feature maps. Here, m_{fuse} represents the ASC reconstruction image by fusing feature maps of different layers, and $m_1 \sim m_6$ represents the feature maps at different stages of the decoders. From the visualization results, the resolution of the feature maps gradually increases from m_1 to m_6, and the local scattering features of the targets become more prominent. The background regions show no activation features, and the clutter is effectively suppressed. This indicates that the proposed model can extract SAR target scattering features and has certain interpretability. In terms of the detection results, the ASC features extracted by the network demonstrate good performance in target localization.

I. Ablation Experiment

The ablative experiments are designed to illustrate the superiority of the proposed ASC-U^{2}Det objectively and comprehensively. Table VI records the impact of the subaperture overlap data enhancement method and the ASC reconstruction branch on the SAR target detection performance. The test set of the experiment consists of SAR images with the single look of 10° subaperture accumulation angle and 8° overlap angle, which is consistent with the configuration of EXP 2.

According to Table VII, both the subaperture overlap data enhancement method and the ASC reconstruction branch are beneficial for improving the target detection performance. First, only expanding the training dataset with subaperture overlap, the model evaluation metrics P, R, F1, and AP are improved by 9.21%, 7.49%, 8.35%, and 10.11%, respectively. Similarly, only adding the ASC reconstruction branch also leads to improvements of 11.85%, 13.09%, 12.46%, and 13.84% in the evaluation metrics. Furthermore, when both methods are used simultaneously, the detection performance is further enhanced, with evaluation metrics reaching 93.22%, 97.48%, 95.31%, and 98.37%. The ablative experiment confirms the effectiveness of the proposed methods.

TABLE VII
RESULTS OF THE ABLATION EXPERIMENT

Sub-aperture overlap	ASC reconstruction branch	P	R	F1	AP
✗	✗	79.94	80.94	80.44	81.81
✓	✗	89.15	88.43	88.79	91.92
✗	✓	91.79	94.03	92.90	95.65
✓	✓	**93.22**	**97.48**	**95.31**	**98.37**

V. CONCLUSION

This article proposes an omnidirectional ASC guided network ASC-U^{2}Det for SAR target detection. First, this article designs a subaperture division data enhancement method and constructs the SAR-Vehicle-Det dataset to support the evaluation of model algorithms. Second, the ASC reconstruction mask maps are constructed as the supervised information to guide the network to extract SAR target scattering features while suppressing background clutter. Numerous experimental results demonstrate that the proposed model has better detection performance compared to existing target detection algorithms. Finally, this article also explores the impact of different subaperture accumulation angles and multilook imaging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

methods on the model performance, providing a reference for the practical application of the algorithm. Future research will further explore the combination of SAR imaging and image interpretation algorithms.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their very competent comments and helpful suggestions.

REFERENCES

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] S. Song, Y. Dai, S. Sun, and T. Jin, "Efficient image reconstruction methods based on structured sparsity for short-range radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024, Art. no. 5212615, doi: 10.1109/TGRS.2024.3404626.

[3] Y. Zhu, J. Ai, L. Wu, D. Guo, W. Jia, and R. Hong, "An active multi-target domain adaptation strategy: Progressive class prototype rectification," *IEEE Trans. Multimedia*, early access, Dec. 24, 2024, doi: 10.1109/TMM.2024.3521740.

[4] J. Pei, Y. Huang, W. Huo, Y. Zhang, J. Yang, and T.-S. Yeo, "SAR automatic target recognition based on multiview deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2196–2210, Apr. 2018.

[5] W. Li et al., "Predicting gradient is better: Exploring self-supervised learning for SAR ATR with a joint-embedding predictive architecture," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 326–338, Dec. 2024.

[6] S. M. Scarborough et al., "A challenge problem for SAR-based GMTI in urban environments," *Proc. SPIE*, vol. 7337, pp. 143–152, May 2009.

[7] M. Duquenoy, J. P. Ovarlez, L. Ferro-Famil, E. Pottier, and L. Vignaud, "Scatterers characterisation in radar imaging using joint time–frequency analysis and polarimetric coherent decompositions," *IET Radar, Sonar Navigat.*, vol. 4, no. 3, pp. 384–402, Jun. 2010.

[8] O. Ponce et al., "Fully polarimetric high-resolution 3-D imaging with circular SAR at L-band," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3074–3090, Jun. 2014.

[9] D. Xiang, W. Li, X. Sun, H. Wang, and Y. Su, "Sidelobe suppression for high-resolution SAR imagery based on spectral reshaping and feature statistical difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, Art. no. 5211314, doi: 10.1109/TGRS.2024.3394405.

[10] F. Teng, W. Hong, and Y. Lin, "Aspect entropy extraction using circular SAR data and scattering anisotropy analysis," *Sensors*, vol. 19, no. 2, p. 346, Jan. 2019.

[11] F. Xue et al., "Analysis of azimuthal variations using multi-aperture polarimetric entropy with circular SAR images," *Remote Sens.*, vol. 10, no. 1, p. 123, Jan. 2018.

[12] J. S. Lee, M. R. Grunes, and R. Kwok, "Classification of multi-look polarimetric SAR imagery based on complex Wishart distribution," *Int. J. Remote Sens.*, vol. 15, no. 11, pp. 2299–2311, Mar. 1994.

[13] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[14] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[15] S. Xian, W. Zhirui, and S. Yuanrui, "AIR-SARShip-1.0: High-resolution SAR ship detection dataset," *J. Radars*, vol. 8, no. 6, pp. 852–862, 2019.

[16] W. Li, W. Yang, Y. Hou, L. Liu, Y. Liu, and X. Li, "SARATR-X: Towards building a foundation model for SAR target recognition," *IEEE Trans. Image Process.*, vol. 34, pp. 869–884, 2025.

[17] (Jan. 19, 2024). *Complex SAR Data*. [Online]. Available: https://www.sandia.gov/radar/pathfinder-radar-isr-and-synthetic-aperture-radar-sar-systems/complex-data/

[18] F. Gao, Y. Yang, J. Wang, J. Sun, E. Yang, and H. Zhou, "A deep convolutional generative adversarial networks (DCGANs)-based semi-supervised method for object recognition in synthetic aperture radar (SAR) images," *Remote Sens.*, vol. 10, no. 6, p. 846, May 2018.

[19] D. Malmgren-Hansen, A. Kusk, J. Dall, A. A. Nielsen, R. Engholm, and H. Skriver, "Improving SAR automatic target recognition models with transfer learning from simulated data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1484–1488, Sep. 2017.

[20] X. Sun, X. Li, D. Xiang, and C. Hu, "SAR vehicle image generation with integrated deep imaging geometric information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104028.

[21] Z. Liu, S. Luo, and Y. Wang, "Mix MSTAR: A synthetic benchmark dataset for multi-class rotation vehicle detection in large-scale SAR images," *Remote Sens.*, vol. 15, no. 18, p. 4558, Sep. 2023.

[22] Z. Wang, L. Du, J. Mao, B. Liu, and D. Yang, "SAR target detection based on SSD with data augmentation and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 150–154, Jan. 2019.

[23] R. Wang, Z. Wang, Y. Chen, H. Kang, F. Luo, and Y. Liu, "Target recognition in SAR images using complex-valued network guided with sub-aperture decomposition," *Remote Sens.*, vol. 15, no. 16, p. 4031, Aug. 2023.

[24] Z. Wang, X. Fu, and K. Xia, "Target classification for single-channel SAR images based on transfer learning with subaperture decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] G.-C. Sun, Y. Liu, M. Xing, S. Wang, L. Guo, and J. Yang, "A real-time imaging algorithm based on sub-aperture CS-dechirp for GF3-SAR data," *Sensors*, vol. 18, no. 8, p. 2562, Aug. 2018.

[26] D. Feng, D. An, L. Chen, and X. Huang, "Holographic SAR tomography 3-D reconstruction based on iterative adaptive approach and generalized likelihood ratio test," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 305–315, Jan. 2021.

[27] H. Wan et al., "AFSar: An anchor-free SAR target detection algorithm based on multiscale enhancement representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5219514, doi: 10.1109/TGRS.2021.3137817.

[28] L. Du, L. Li, D. Wei, and J. Mao, "Saliency-guided single shot multibox detector for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3366–3376, May 2020.

[29] W. Li, W. Yang, W. Zhang, T. Liu, Y. Liu, and L. Liu, "Hierarchical disentanglement-alignment network for robust SAR vehicle recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9661–9679, 2023.

[30] Z. Huang, Y. Liu, X. Yao, J. Ren, and J. Han, "Uncertainty exploration: Toward explainable SAR target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2023, Art. no. 4202314.

[31] S. Feng, K. Ji, F. Wang, L. Zhang, X. Ma, and G. Kuang, "PAN: Part attention network integrating electromagnetic characteristics for interpretable SAR vehicle target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, Art. no. 5204617, doi: 10.1109/TGRS.2023.3256399.

[32] Z. Liu, L. Wang, Z. Wen, K. Li, and Q. Pan, "Multilevel scattering center and deep feature fusion learning framework for SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5227914, doi: 10.1109/TGRS.2022.3174703.

[33] Y. Li, L. Du, and D. Wei, "Multiscale CNN based on component analysis for SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, Art. no. 5211212, doi: 10.1109/TGRS.2021.3100137.

[34] Y. Guo, Y. Liu, T. Georgiou, and M. S. Lew, "A review of semantic segmentation using deep neural networks," *Int. J. Multimedia Inf. Retr.*, vol. 7, no. 2, pp. 87–93, 2018.

[35] M. Zhao, X. Zhang, and A. Kaup, "Multitask learning for SAR ship detection with Gaussian-mask joint segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5214516, doi: 10.1109/TGRS.2023.3304847.

[36] J. Wang, W. Yang, H.-C. Li, H. Zhang, and G.-S. Xia, "Learning center probability map for detecting objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4307–4323, May 2021.

[37] C. H. Casteel, L. A. Gorham, M. J. Minardi, S. Scarborough, K. D. Naidu, and U. Majumder, "A challenge problem for 2D/3D imaging of targets from a volumetric data set in an urban environment," *Proc. SPIE*, vol. 6568, Apr. 2007, Art. no. 65680D.

[38] L. Chen, D. An, and X. Huang, "A backprojection-based imaging for circular synthetic aperture radar," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3547–3555, Aug. 2017.

[39] G. Jia, M. F. Buchroithner, W. Chang, and Z. Liu, "Fourier-based 2-D imaging algorithm for circular synthetic aperture radar: Analysis and application," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 1, pp. 475–489, Jan. 2016.

[40] G. Jia, W. Chang, Q. Zhang, and X. Luan, "The analysis and realization of motion compensation for circular synthetic aperture radar data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 3060–3071, Jul. 2016.

[41] K. E. Dungan and J. W. Nehrbass, "Wide-area wide-angle SAR focusing," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 29, no. 1, pp. 21–28, Jan. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.

[42] M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jun. 2014.
[43] M. J. Gerry, L. C. Potter, I. J. Gupta, and A. Van Der Merwe, "A parametric model for synthetic aperture radar measurements," *IEEE Trans. Antennas Propag.*, vol. 47, no. 7, pp. 1179–1188, Jul. 1999.
[44] M. Jing and G. Zhang, "Attributed scattering center extraction with genetic algorithm," *IEEE Trans. Antennas Propag.*, vol. 69, no. 5, pp. 2810–2819, May 2021.
[45] B. Ding, G. Wen, X. Huang, C. Ma, and X. Yang, "Data augmentation by multilevel reconstruction using attributed scattering center for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 979–983, Jun. 2017.
[46] H. Liu, B. Jiu, F. Li, and Y. Wang, "Attributed scattering center extraction algorithm based on sparse representation with dictionary refinement," *IEEE Trans. Antennas Propag.*, vol. 65, no. 5, pp. 2604–2614, May 2017.
[47] H.-C. Chiang, R. L. Moses, and L. C. Potter, "Model-based classification of radar images," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1842–1854, Aug. 2000.
[48] Y. Zhuge, Y. Zeng, and H. Lu, "Deep embedding features for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, Jul. 2019, pp. 9340–9347.
[49] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, "U^2-Net: Going deeper with nested U-structure for salient object detection," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107404.
[50] Y. Zhang and Q. Yang, "A survey on multi-task learning," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 12, pp. 5586–5609, Dec. 2022.
[51] C. Zhou, "Computation of optical flow using a neural network," in *Proc. IEEE Int. Conf. Neural Netw.*, vol. 2, Jun. 1988, pp. 71–78.
[52] P. Wang et al., "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1451–1460.
[53] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, Apr. 2020, pp. 12993–13000.
[54] J. V. Bouvrie, "Notes on convolutional neural networks," Dept. Center Biol. Comput. Learn., Massachusetts Inst. Technol., Cambridge, MA, USA, Tech. Rep., 2006, pp. 38–44.
[55] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[56] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[57] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.
[58] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.
[59] J. Zhou et al., "DiffDet4SAR: Diffusion-based aircraft target detection network for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[60] Y. Li et al., "SARDet-100K: Towards open-source benchmark and toolkit for large-scale SAR object detection," in *Proc. The 38th Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2024, pp. 1–12.
[61] Y. Dai, M. Zou, Y. Li, X. Li, K. Ni, and J. Yang, "DenoDet: Attention as deformable multi-subspace feature denoising for target detection in SAR images," *IEEE Trans. Aerosp. Electron. Syst.*, pp. 1–17, Nov. 2024, doi: 10.1109/TAES.2024.3507786.
[62] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3912–3921.
[63] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.



Di Wang received the B.S. degree in electronic information science and technology from Hefei University of Technology, Hefei, China, in 2021. He is currently pursuing the Ph.D. degree in information and communication engineering with the National University of Defense Technology, Changsha, China.

His research interests include SAR image interpretation, SAR target detection, and recognition.



Yongping Song received the B.S. degree in electronic engineering and the M.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2012, 2014, and 2019, respectively.

He was a Lecturer at Air Force Early Warning Academy until 2021, and now is an Assistant Researcher of National University of Defense Technology. His research interests include MIMO radar image formation, radar targets detection, radar anti-jamming.



Leping Chen received the B.S. degree in electronic engineering and the M.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2011, 2014, and 2018, respectively.

He is currently a Lecturer with the National University of Defense Technology. His current research interests include circular synthetic aperture radar image formation and high-resolution SAR image formation.



Daoxiang An (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2004, 2006, and 2011, respectively.

He is currently an Associate Professor with the National University of Defense Technology. His current research interests include ultra-wideband SAR, circular SAR, video SAR imaging, SAR interferometry, SAR-GMTI, and SAR image interpretation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:06:41 UTC from IEEE Xplore. Restrictions apply.